make old XCF files look exactly as before) and default (mostly linear).

The new blend modes are:

- LCH layer modes—Hue, Chroma, Colour, and Lightness
- Pass-Through mode for layer groups
- Linear Burn, Vivid Light, Linear Light, Pin Light, Hard Mix, Exclusion, Merge and Split

Layers, paths and channels can also be tagged with colour labels to improve project organisation. Compositing options for layers are exposed to users now, and all layer-related settings are finally available in the *Layer Attributes* dialogue box.

Moreover, if you always need alpha in your layers, you can enable the automatic generation of the alpha channel in imported images upon opening them. See *Edit > Preferences > Image Import and Export page* for this and more policies. Layer groups can finally put on masks.

Colour dialogue boxes now have an LCH colour selector that you can use instead of HSV. The LCH selector also displays out-of-gamut warnings. A new Hue-Chroma filter in the *Colours* menu works much like *Hue-Saturation* but operates in the CIE LCH colour space. The *Fuzzy Select* and *Bucket Fill* tools can now select colours by their values in CIE L, C, and H channels. Both the *Colour Picker* and the *Sample Points* dialogue boxes now display pixel values in CIE LAB and CIE LCH as per your preference.

The *New Unified Transform* tool simplifies making multiple transforms, such as scaling, rotating, and correcting perspective in one go.

The new *Warp Transform* tool allows you to do localised transforms like growing or shifting pixels with a soft brush and undo support. As such, the new tool replaces the old iWarp filter that was innovative at the time of its inception (and pre-dated Photoshop's Liquify filter), but was ultimately cumbersome to use. The *Warp Transform* tool also features an *Eraser* mode to selectively remove changes, previously unavailable in the iWarp filter.

The new *Handle Transform* tool provides an interesting approach at applying scaling, rotating and perspective correction using handles placed on the canvas. People who are used to editing on touch surfaces might find this tool surprisingly easy to grasp.

The *Blend* tool has been renamed the *Gradient* tool and its default shortcut changed to G. But this pales in comparison to what the tool can actually do now, which is a lot. The new tool pretty much makes the old *Gradient Editor* dialogue box obsolete. Now, you can create and delete colour stops, select and shift them, assign colours to colour stops, change blending and colouring for segments between colour stops, and create new colour stops from midpoints right on the canvas.

All gradients available by default are also editable now, which means that when you try to change an existing gradient from a system folder, the GIMP will create a copy of it, call it a custom gradient and preserve it across sessions—unless, of course, you edit another 'system' gradient, in which case it will become the new custom gradient.

The *Foreground Select* tool can finally make sub-pixel selections in complex cases such as strands of hair on a textured background. Two new masking methods are now available for that.

Both the *Select by Colour* and *Fuzzy Select* tools now feature a *Draw mask* option to display future selection areas with a magenta fill. The latter tool also has a *Diagonal neighbours* option to select diagonally neighbouring pixels.

For the *Free Select* tool, closing a polygonal/free selection now does not confirm the selection automatically. Instead, one can still tweak positions of nodes (where applicable), then press *Enter*, and double-click inside the selection or switch to another tool to confirm the selection.

The *Intelligent Scissors* tool finally allows you to remove the last added segment with the *Backspace* key, and the GIMP now checks whether the first and the last segments are distinct before closing the curve.

All colour tools have been refactored to become GEGL based filters. Hence, the *Colour* sub-menu in the *Tools* menu has been removed, and these filters are now mostly unavailable in the toolbox.

The *Text* tool now fully supports advanced input methods for CJK (Chinese, Japanese, Korean) and other non-Western languages. The pre-edit text is now displayed just as expected, depending on your platform and the input method engine. Several input method-related bugs and crashes have also been fixed.

The *N-Point Deformation* tool introduces the kind of smooth (and as little rigid as possible) warping one would expect physical objects to have.

The *Seamless Clone* tool is aimed at simplifying the making of layered compositions. Typically, when one pastes one image into another, there are all sorts of mismatches—in colour temperature, brightness, etc. This new experimental tool tries to adapt various properties of a pasted image to its backdrop.

To enable these tools, you need to first enable the *Playground* page of the *Preferences* dialogue. This is done by running GIMP with a '—show-playground.' switch (for Windows, one might want to tweak the path to the GIMP in the shortcut properties, accordingly). Then go to *Edit -> Preferences -> Playground* and enable the respective options so that the tools show up in the toolbox.

In this article, we have explored just the basic tools of the GIMP. Stay tuned for an article on more advanced tools soon. END

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